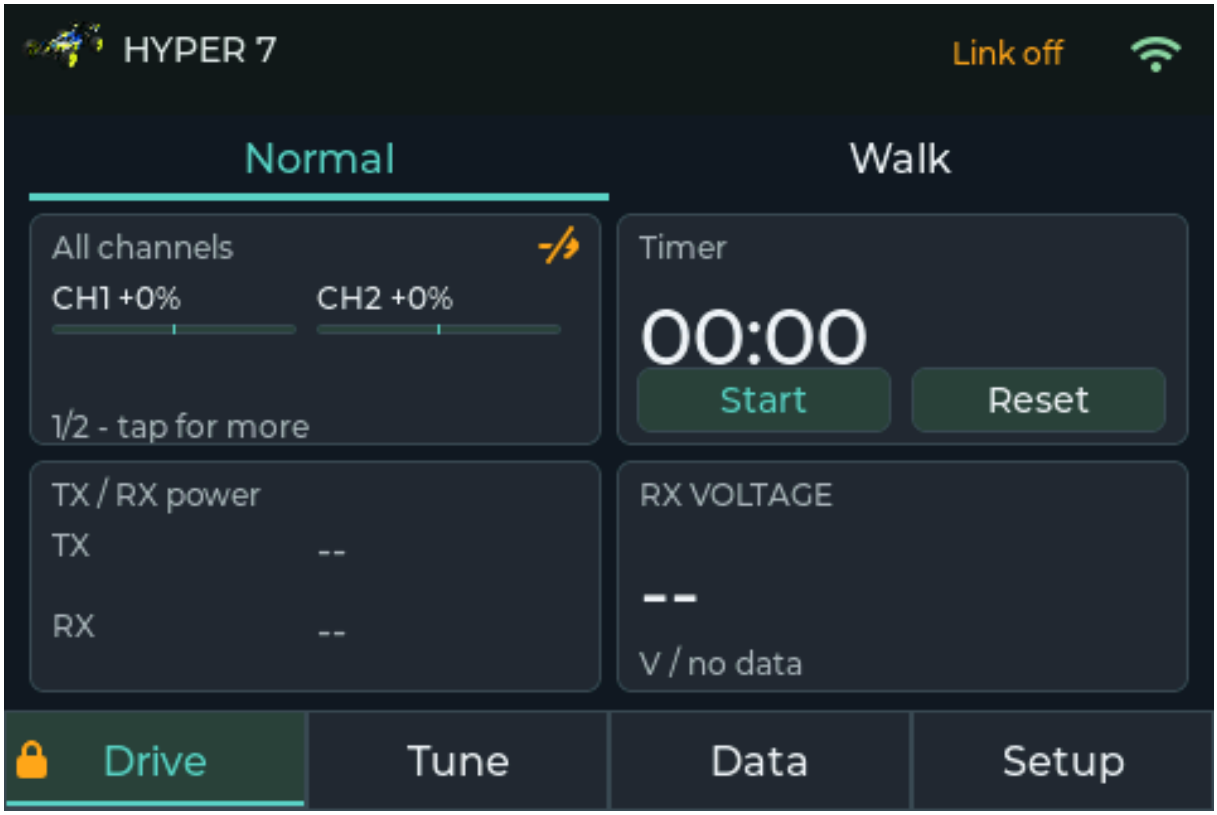
JR R-1 EVO

User manual

Prototype edition 1.3
1 November 2026
Based on verified firmware 06b6928



Original JR controls. A new touchscreen interface. ExpressLRS radio and telemetry.

Operating guide for the custom XIAO ESP32-S3 / Waveshare LCD (F) transmitter. Includes channel setup, mixing, timers, maps, web configuration and safety states.

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1. Quick start

Before using a model

Confirm the selected model, channel directions, neutral, travel and receiver/ESC failsafe before driving. For first setup, disconnect motor power or otherwise make the vehicle unable to move. CUT is not a replacement for receiver failsafe.

The transmitter and RF module may have separate power feeds in this conversion. USB power can run the screen without powering the Ranger or producing a detected TX battery voltage. Follow the installed wiring; do not assume a lit screen means the RF module is powered.

Start a normal driving session

1. Power the handset and enter its four-digit PIN. The trial default is `1234`; the correct fourth digit submits automatically.

2. Confirm the model name and artwork in the title bar. Select the correct model before enabling output.

3. When prompted after TX battery detection, approve **module communication** on the handset if you intend to use the radio. Cancel leaves it paused.

4. Power the receiver and confirm a fresh link indication. A high LQ alone does not release CUT.

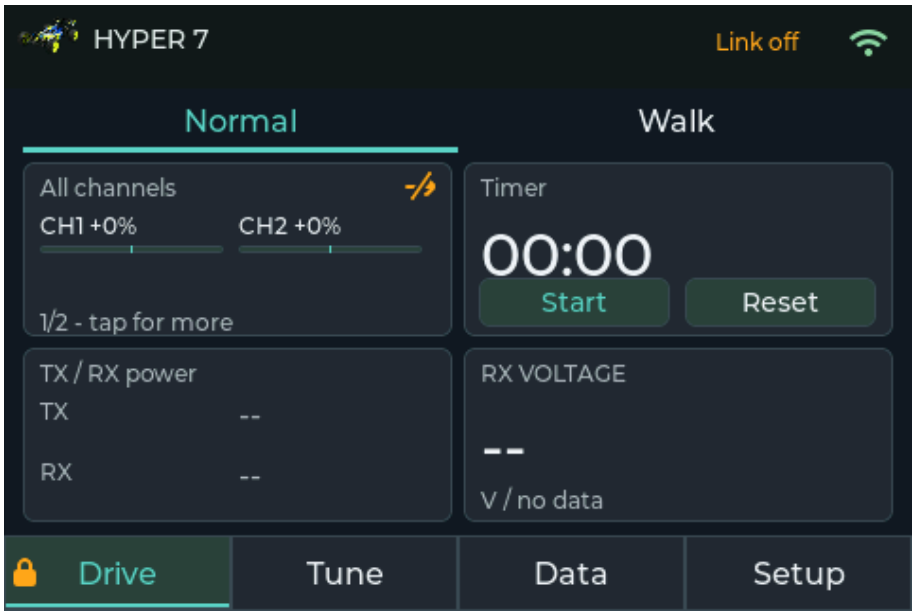
5. Select **Normal** or **Walk** on Drive. Centre the throttle.

6. Hold the **Drive** navigation control deliberately. Observe the progress or readiness message, then release as prompted. If output remains CUT, use **Output status** to see the reason.

7. Verify steering direction and throttle/brake operation before driving.

Finish a session

Hold Drive to return to CUT and check its state indicator. Secure/power down the vehicle before handling it. Switch off module power when RF silence is required. Lap records are held in RAM: export any records you want to keep before turning off or restarting the handset.



Native handset Drive screen

Most screens in this guide are native rendered captures from the XIAO at firmware 7d13dbf. Widget-focus examples are separately labelled simulator captures at 06b6928. Earlier native captures may differ slightly from the current timer or widget layout. Neither capture method reads back the physical LCD.

2. Controls and navigation

The four main sections

Section	Purpose
Drive	Widgets, operating mode and deliberate CUT/output-enable action.
Tune	Quick steering, throttle and brake response adjustments.
Data	Sensors, GPS map, channels and physical input monitoring.
Setup	Model configuration and handset-wide settings.

Tap the bottom navigation control to change section. An ordinary navigation tap does not enable output. Detail pages provide Back navigation; use their displayed controls to return to the parent page. Apply/Save and Cancel matter in editors: leaving an editor is not a substitute for applying its draft.

Where page dots are shown, tap a dot or swipe horizontally to change page. Tabs select a category within the current section. Paging and tab selection are separate operations.

The title bar identifies the current model and link/network status. Tap the model area to reach model selection. Tap the Wi-Fi icon for Wi-Fi configuration. Hold the LQ/status area for the module-communication dialog.

Original programming keypad

Key	Current navigation action
MODE	Open Quick menu; hold to return to Drive, except where a modal editor requires finishing/cancelling.
PAGE	Next page; hold for previous page.
UP / DOWN	Move focus between selectable items; held keys repeat.
SELECT / SEL	Activate the focused item.
+ / -	Adjust the selected editor value; outside editors, select adjacent tabs where supported.
CLR / CLEAR	Back; hold in a value editor to request resetting that draft field. Apply is still required.

From Quick menu, UP opens steering expo, PAGE endpoints, SEL throttle, DOWN trims, + models, - timer, CLR Drive and MODE Setup. A first key press used to wake the screen is consumed rather than also activating a control.

Current limitation: the eight programming keys are reserved by navigation. Although they appear in the assignment catalogue, assigning MODE or another reserved key as a general channel control does not currently override navigation. The wheel, trigger, trims and other supported switches use the assignment

system. The original DISPLAY switch is not wired as a XIAO input in this build.

3. CUT, module communication and PIN

These are different states:

State	What it controls
PIN locked	Access to the handset interface. Unlocking does not enable output.
CUT	Live requested channel values are inhibited; safe values are used.
Module communication paused	The XIAO stops CRSF channel and management traffic to the Ranger.
RF module unpowered	Hardware power is removed from the Ranger. This is the means of guaranteeing it is off.

CUT does not mean RF off. With module communication enabled, safe CRSF channel frames continue during CUT. Pausing CRSF has not yet been validated as guaranteed RF silence on the installed Ranger; use its power switch/feed when that matters.

The handset restarts in CUT with module communication paused. A stable sensed TX battery feed prompts for physical approval to enable communication. Approval does not arm the outputs. Rebooting or losing the sensed supply revokes that approval; it is not remembered for the next power session. TX voltage sensing must be On.

Releasing CUT

Hold Drive, or open **Setup > Model > Drive modes > Output status** and hold **Enable output**. The readiness explanation is authoritative. Typical blockers are neutral required, Link wait, invalid inputs/configuration, startup recovery, a paused module link or an active setup session.

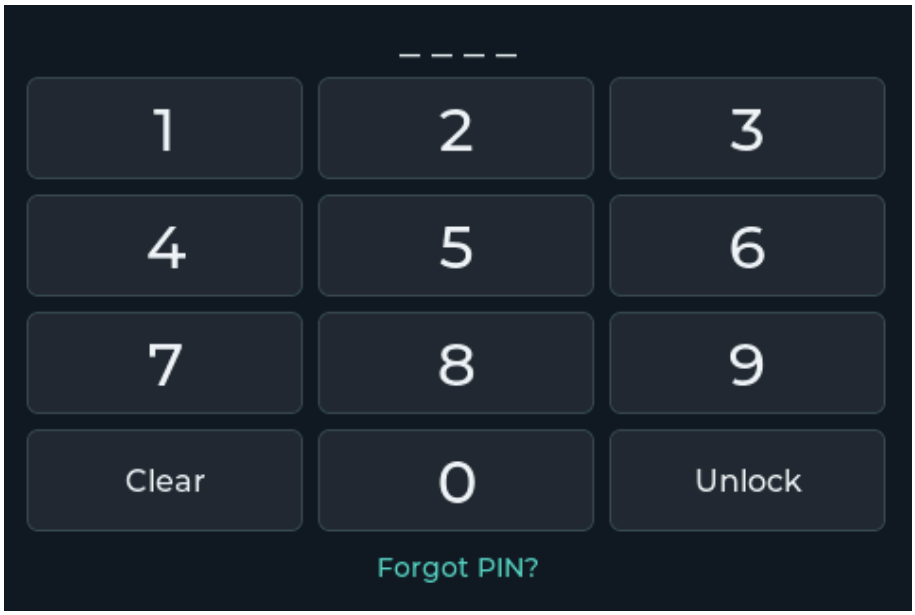
If the message says **Setup session active: close it on the handset to arm**, use **Finish Wi-Fi setup** on Output status. Merely closing a browser tab may not end the handset's setup session. You do not need to erase Wi-Fi credentials.

Structural changes such as channel/mix edits, model changes and calibration can force CUT. Recheck the result and deliberately re-enable at neutral afterward. Loss of return telemetry after arming is currently advisory; actual radio-loss behaviour is governed by the receiver/ESC failsafe.

PIN change and recovery

Open **Setup > Handset > System > PIN** for change/recovery options. Follow the prompts to enter the existing and replacement four-digit PIN. **Forgot PIN?** opens USB recovery instructions.

For recovery, connect the XIAO USB data port, open a 115200-baud serial terminal, and send `PINRESET 1234` followed by Enter, substituting your new four digits if wanted. Outputs must be CUT. Read the success response. Recovery leaves outputs CUT and does not cancel Walk mode. Physical USB access is therefore a recovery path, not a security boundary against someone with access to the handset.



PIN entry

4. Models

Tap the title-bar model area to open the model list. A row marked Load selects that memory; the active row is marked Active. Load a model in CUT with throttle centred, then verify the active name. Use the separate **Appearance** control to change the active model's image and paint. Rename the active model in the web UI (1-12 printable ASCII characters). Do not identify a vehicle by image alone: multiple memories can use the same artwork.

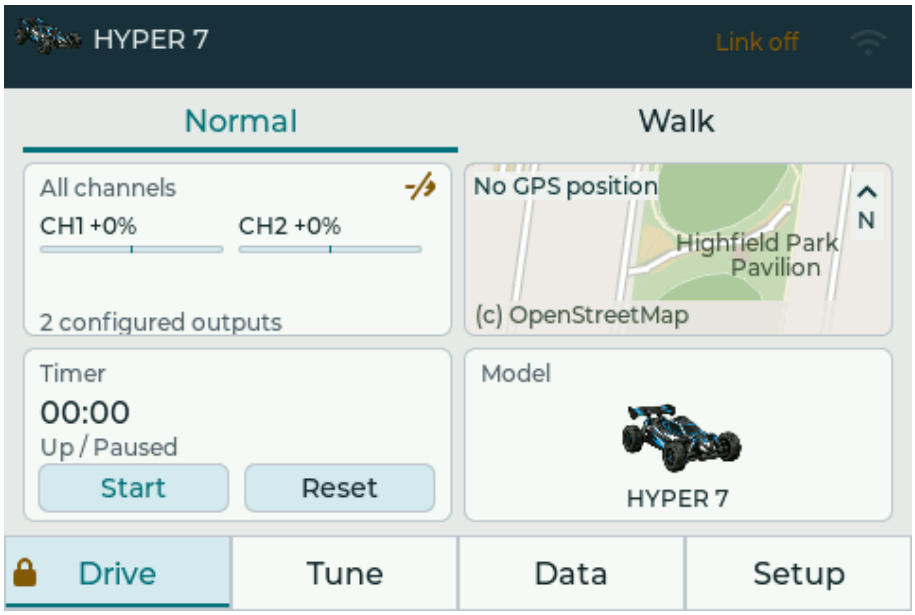
Artwork choices include Hyper 7, buggy, truggy, short-course truck, monster truck, crawler, touring car and Formula/F1. Primary, accent and base paint affect separate image regions; tyres and mechanical shading retain their appearance. The title-bar icon uses the selected artwork too.

Model settings include channel response/mixes, assignments, dashboard, drive-mode settings and timers. Display and network settings belong to the handset. Changing model is not receiver binding and does not automatically prove the selected receiver/vehicle matches it. Check Model Match/binding separately if used.

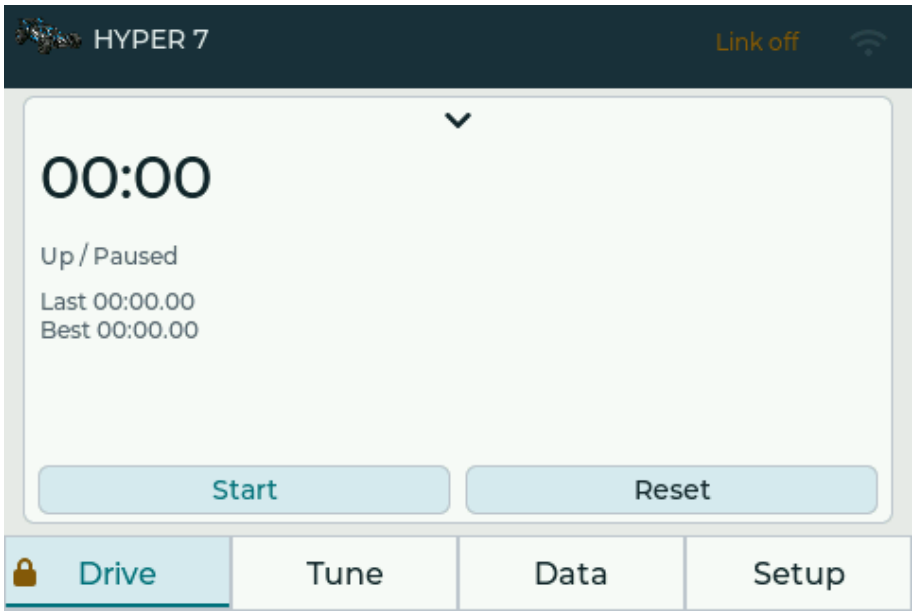
5. Drive dashboard and widgets

Open **Setup > Model > Dashboard**, select a position and edit its type, size, style and source. Apply the change. The grid has two columns and two rows: **1×1**, **2×1**, **1×2** and **2×2** are available where they fit. Enlarging a widget covers neighbouring positions without deleting their saved widgets; reducing it reveals them again.

Hold the widget body for about 0.65 seconds and release to maximise it across the dashboard. The title bar and bottom navigation remain available. Use the small down-chevron overlaid at the top centre, Back, or another long press to return to the grid. Short taps still open details or page through entries. Timer Start/Reset buttons keep their own actions; hold the timer body above them to enlarge it. The saved layout is unchanged. The web mirror supports the same gesture. The expanded widget omits its own title and extra Restore row to leave more room for data.



Simulator example: saved dashboard before expansion; simulated inputs, not a physical LCD capture.



Simulator example: Timer expanded, with the overlaid down-chevron returning to the unchanged dashboard.

Widget	Use and options
Timer	Stopwatch, countdown or lap display. Controls and larger layouts expose session information; tap the title/time area for timer pages.
TX / RX power	Local transmitter battery and receiver-supply telemetry. Numeric, bar or dial styles where offered.
Channels	One selected channel, optionally a second. Select a suitable numeric, bar, dial, graph or switch style.
All channels	Configured outputs, physical controls or both. Choose visible entries; horizontal/vertical bars or numbers, with binary indicators for suitable switch states. Tap to page through entries.

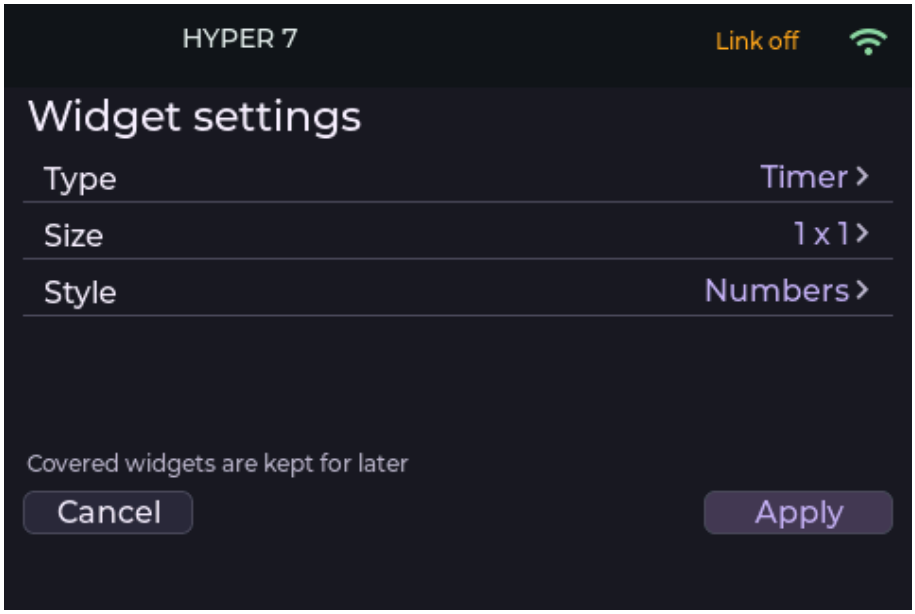
Widget	Use and options
Sensor	Speed, temperature, RPM, RX voltage, LQ, satellites, drive-pack total, lowest cell, FAS voltage or current. Styles depend on the sensor.
Battery cells	Individual reported cells together, plus total, lowest cell and count. Auto or expected 1S-12S; page when needed.
GPS map / trail	Saved map with north indicator, attribution, GPS position and trail. Tap for the larger GPS view.
Model image	Active model name and painted vehicle image. Tap to reach model selection.
Empty	Leave the position unused.

Speed supports a dial and current/max/average readings; it is not calculated from throttle position. Tap a sensor for detail/history or power for radio diagnostics.

Monitoring controls while CUT

Channel widgets show requested control values even while CUT sends safe values. A small crossed-out output indicator marks that condition. Use **Data > Channels** to distinguish requested values from actual sent values. Moving bars during CUT are useful for checking wiring/calibration; they do not mean the servo is moving.

In All channels, choose whether to include hardware controls and which entries are visible. Timer and grip A/B inputs can be shown without assigning an RF channel. Binary indicators are status displays, not commands to operate switches.



Widget editor

6. Channel setup and tuning

Open **Setup > Model > Channels**, page to CH1-16 and select the channel. It has three groups: **Role / physical input**, **Response / trim** and **Mixing / output limits**.

Role, input and calibration

Steering and throttle roles can be moved to suitable free channels. CH1 steering and CH2 throttle are conventional starting points, not permanent requirements. An occupied/conflicting channel is rejected. Auxiliary controls are assigned through Control map; a role's Physical control entry links to its assignments.

Use Calibration / deadband for supported proportional inputs. Follow the on-screen sequence, moving through the full usable range and returning to the true centre. Save only after checking the neutral reading. Calibration describes the wheel or trigger electrically; endpoints limit servo output. Changing endpoints does not repair a disconnected or incorrectly calibrated potentiometer.

Reverse steering or another channel

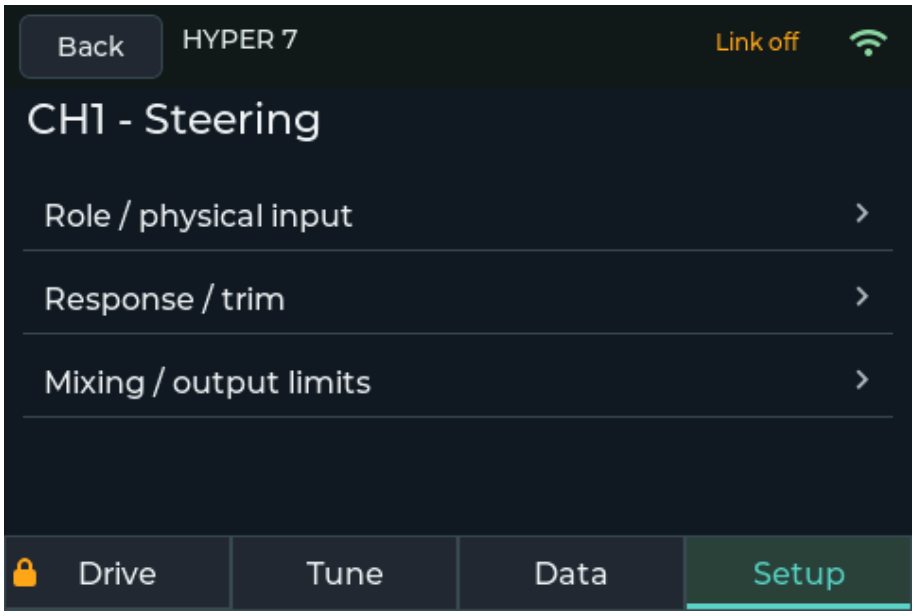
1. Open **Tune**, choose Steering or the relevant response page.
2. Go to the page containing **Direction** and change Normal to Reversed.
3. Apply if the editor presents an Apply action.
4. Test at small travel, then verify both endpoints and neutral again.

For an auxiliary channel, use its Response / trim > Quick adjustments. Reverse one intended stage; reversing both a control assignment and its channel can cancel the intended effect.

Response settings

Setting	Effect
Rate	Scales travel/sensitivity for the selected side.
Expo	Shapes response around centre while retaining the selected travel limits. Inspect the curve rather than assuming another radio's sign convention.
Endpoint	Maximum negative/positive travel; steering shows physical left/right values.
Response speed	Slows changes over the configured time in milliseconds.
Direction	Reverses output direction.
Trim	Small operational centre adjustment.
Subtrim	Centre adjustment used during setup.
Response offset	Detailed response adjustment where offered.

Tune provides quick adjustments and **Channel setup** links to fuller settings. Detailed response separates Right/Left, Forward/Brake or Positive/Negative. Trim control and Subtrim control entries can select the physical adjuster directly. After moving a centre or reversing a channel, recheck mechanical travel for binding.



Channel configuration

7. Hardware-control assignments

Open **Setup > Model > Control map**, or follow a channel's Physical control link. Select the intended physical control, enable its assignment, select the target and set its behaviour. Save the assignment. Availability depends on the control and target; unsupported combinations are rejected.

Setting	Meaning
Target	CRSF channel, supported transmitter action, model parameter or drive-mode function.
Mode	Proportional, momentary, toggle, two-position, stepped, three-position or pulse where supported.
Range / step	Minimum/maximum values and size of each step.
Initial / safe state	Starting and safe values where the selected mode supports them.
Reverse	Reverse this assignment's control sense.
Pulse duration	Duration of a supported pulse action.
Context	Restrict where/when the assignment is active.

Context **Always** applies regardless of screen; **Dashboard** applies on Drive; **Menus** applies away from the dashboard; **CUT only** applies while output is inhibited; **Armed only** applies during enabled output. Context does not bypass CUT or make an unsafe output live. A disabled assignment can fall back to the control's normal action; it is not always the same as disabling the physical input.

A maintained two-position switch is suited to a binary auxiliary output. A spring-return trim can step a value or adjust a parameter. Check the live input monitor first, then check requested and sent channel values. Saved memories may retain older assignments, so inspect them instead of assuming current defaults.

8. Mixing and switched rates

Open the **destination** channel, then **Mixing / output limits**. The destination is the receiver channel you want to change. The source is the input/channel whose movement should influence it. Up to two mix contributions are available.

The Sections menu provides **Mix sources**, **Activation**, **Switched rates**, **Baseline / limits** and **Preview / graph**.

Mix setting	Meaning
Enabled / Source	Turn the contribution on and select a raw physical input, channel or configured steering/throttle input.
Positive / negative weight	Scale each side of a proportional source; negative weights invert its contribution.
Add / Replace	Add to the running mix or replace it at that stage.
Follow source trim	Include source trim where supported.
Existing channel input	Include the original channel input at 100%, or use mixes only.
Constant contribution	Fixed baseline before mix contributions.
Final output minimum / maximum	Bound the mixed output.
Aux CUT override / value	Auxiliary safe output during CUT; unavailable for steering/throttle roles.

Activation can be Always, Switch low/high, Walk or Normal. Source mode can be Proportional or Switch above/below a threshold. Hysteresis prevents repeated switching near the threshold. Threshold switch modes use their defined switched contribution; do not expect proportional weights to behave as a gain curve there.

Switched rates use a selected physical switch, separate low/high negative and positive rates, and a transition time. This is one way to select reduced versus full authority without repeatedly editing the ordinary endpoint settings.

Example: reduce CH3 gyro gain as forward throttle rises

This example describes a **gain-channel mix**, not a gyro implemented in the handset. First establish which CH3 values give the required mode/gain on your actual gyro. Its centre may divide rate and heading-hold modes; crossing centre can change mode rather than simply reduce gain.

For a preview example, choose a baseline of +70% and reduce it to +30% at full forward throttle:

1. Select CH3 as the destination; avoid an unintended additional CH3 assignment.

2. Set Existing channel input to None (mixes only).

3. Set Constant contribution to +70%.

4. Enable Mix 1 with configured Throttle input as source, Add operation, proportional source mode, positive weight -40%, negative weight 0% and no source trim. Leave Mix 2 disabled.

5. Set final output limits to +30% and +70%.

6. In preview, check neutral gives +70%, half forward gives about +50%, full forward gives +30%, and brake leaves the baseline unchanged.

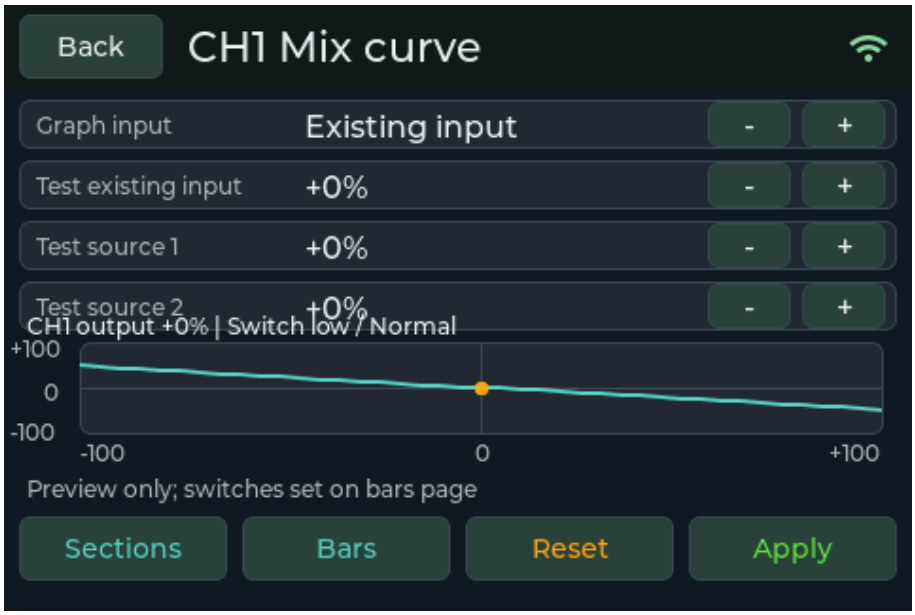
7. Apply and verify the real gyro's behaviour with the vehicle immobilised.

Those percentages are illustrative transmitter values, not a calibration or recommended gain for every gyro. Verify the forward source sign in preview.

To make a switch select gyro off/on, configure an appropriate switched auxiliary value or conditional contribution for the same destination, using the gyro's verified off value. Merely disabling this throttle mix leaves its baseline gain active. Do not assume a negative gain value means off.

Preview and resetting

Bars show base/constant, each contribution, mixed and output values. Graph sweeps the selected input while the other test values remain fixed. Test switch and drive mode conditions on the bars page. Preview is simulated and does not transmit the test values. Apply saves; Reset restores the mix draft to defaults and still requires Apply. Self-references/cyclic channel dependencies are rejected.



Mix response graph

9. Normal and Walk modes

Choose Normal or Walk in Drive. Normal uses the model's configured response. Walk limits **physical forward throttle** to the configured percentage, initially 35%, while retaining configured brake/reverse authority. It is a throttle limit, not measured cruise control or a guarantee of a particular vehicle speed.

Set the Walk limit under **Setup > Model > Drive modes**. Mode changes force CUT. Re-enable deliberately with neutral throttle. Leaving Walk requires the PIN; an idle Walk-exit PIN prompt returns to Drive after about 10 seconds and leaves Walk/CUT in force. PIN entry never itself arms output.

10. Stopwatch, countdown and lap timer

Open the Timer widget's title/time area, or press **MODE**, then **-** in Quick menu. Timer has **Controls**, **Laps** and **Settings** tabs. Settings belong to the active model.

Mode	Main action while running
Up	Counts elapsed time; main action pauses, then resumes.
Down	Counts down from Race / countdown duration; continues as +overtime after zero.
Lap	Shows current lap; main action records a lap and starts timing the next one.

In **Down** mode, the buzzer gives one short beep at **20 seconds remaining**, two at **10 seconds**, then one at each of **5, 4, 3, 2 and 1 seconds**. At zero it gives an **800 ms end beep**; the timer continues into overtime. Countdown cues use the selected buzzer pitch and require both Buzzer and Race-end alarm to be On. A short countdown uses only the milestones within its duration. Pausing stops a pending sequence; resuming does not replay passed cues. Reset or Ready starts a new countdown sequence. Sound scheduling does not block the control loop.

In Lap mode, the initial action is Start. Use the separate Pause action when you want to stop timing rather than record a lap. Resume excludes time spent paused. Reset stops/zeros the clock but **keeps recorded laps**. Clear laps is separate, requires confirmation, and is allowed only when paused and not Ready.

Record a race

1. Choose Lap in Settings and set Minimum lap to reject accidental rapid presses (default 3 seconds).
2. Set Race / countdown duration, Race-end alarm and optional Target lap alarm.
3. Start from Controls or the timer's assigned physical button.
4. Press Lap on each crossing. Accepted laps beep; too-short laps are ignored.
5. Pause after the session, review Laps and export before power-off if needed.

Up to **100 laps per model per power session** are held in RAM. Laps shows records and marks the best; summaries include last/best/average/total. At capacity, further laps are not stored until you pause and clear. Settings and accumulated on-time are saved; lap records and running clock are not restored after reboot. On-time is checkpointed about once a minute, so abrupt power loss can lose the latest fraction of that accumulated figure.

Ready: throttle auto-start

Enable Throttle auto-start in Settings, then deliberately press Ready. Centre throttle, then apply forward throttle (about 10% or more) to start timing. Invalid inputs or locked/calibration/recovery conditions prevent a valid start. Ready resets the clock while retaining lap records. Enabling auto-start alone does not put the timer into Ready. **Starting a timer never releases CUT or enables RF.**

A new default timer-button assignment starts/pauses Up/Down and starts/records Lap mode. Existing saved assignments are preserved. Control map also offers supported timer actions such as Lap, pause/resume and Ready. The web timer panel provides settings/actions and lap CSV download; web writes require CUT access.

11. Telemetry and GPS

Use **Data > Sensors** for values and histories, **GPS** for location/trail, **Channels** for requested/sent outputs and **Inputs** for physical controls. No data or stale values mean no usable fresh measurement; they are not zero.

Link data can include LQ, RSSI and SNR. Other supported readings include receiver voltage, speed, GPS/satellites, RPM, temperature, pack/cell values and FAS voltage/current, when supplied by compatible receiver firmware and sensors.

GPS status distinguishes reception from fix quality:

- **Waiting for GPS telemetry:** no usable GPS report is available yet. - **Fix unknown:** fresh usable coordinates have arrived, but standard S.Port GPS does not provide an explicit fix-quality field or satellite count. - **GPS stale / no fix:** position data is absent or older than the freshness window (three seconds). The dashboard can show **No GPS position** before any usable trail point has been accepted. A 0,0 unknown-fix placeholder is not plotted.

Do not treat an empty satellite display as proof that no S.Port GPS data is arriving. Check the position, speed and freshness together.

Which battery is being displayed?

Reading	Measurement
TX	Local handset pack via its voltage-divider input. It is not USB/5V_SYS voltage.
RX	Receiver supply/BEC telemetry.
Drive pack / cells	Vehicle battery measurements from a compatible sensor.
FAS voltage/current	Measurements from the FAS sensor; not automatically the same as RX supply.

Battery cells can use Auto or an expected 1S-12S count. An expected count does not create missing data or divide total voltage into measured cells. The receiver cell trial currently forwards 4S; larger display capacity is not proof of validated 6S/12S receiver support.

The custom receiver adds S.Port support. The GPS-forwarding trial firmware was uploaded to the ER6G on 1 November 2026; live GPS delivery still needs confirmation. A sensor branded RadioMaster/ERS is not necessarily an S.Port sensor; match its protocol and receiver firmware rather than selecting by brand. Receiver wiring/output mapping remains separate from transmitter mixing.

A map can be viewed without GPS; a live position/trail requires valid position telemetry. The GPS page keeps its title bar, navigation and controls visible while idle; automatic fullscreen is disabled for this trial. Use Zoom -/+, Trail and Reset trail as needed. Attribution remains part of the map.

12. Wi-Fi and the web interface

Connect to home Wi-Fi

Open **Setup > Handset > Network / web > Home Wi-Fi**, or tap the Wi-Fi icon. Select/enter the network and password and connect. Password characters are visible in the current setup editor. Saved credentials and optional automatic connection are retained; with auto-connect On, attempts begin after boot without opening setup. The icon is a shortcut/status indication, not proof that every web request works.

Find the actual address and web credentials under **Web access**. Use its numeric IP if the local hostname does not resolve. The installed prototype is currently reserved at `http://192.168.1.251/`; another installation/network may differ. The hostname can be changed through the network/web settings; use the displayed active name/address after applying or restarting.

Hotspot and authentication

If needed, enable the setup hotspot from the LCD and join it from your computer or phone. Read its displayed address/credentials. Home Wi-Fi and hotspot access are different network paths. A browser may need internet access separately to download map imagery/search results.

Web authentication uses username **r1** and the handset's displayed web password. It is separate from the four-digit PIN. Authenticated configuration may be used while the LCD PIN is locked. The live mirror also accepts PIN entry. Do not share a URL containing test login credentials; use the normal login workflow where the client supports it. This guide deliberately includes no installation password.

Live screen and configuration

Open the live handset screen to view the current rendered LCD pixels and operate the menus. The mirror is native 480×320 in landscape; browser zoom can change its apparent size. It is separate from the desktop simulator. Touch/hold gestures should be performed on the mirrored screen as on the LCD.

Configuration and remote writes are restricted by authentication, session checks and CUT state. Physical approval for module communication remains a handset action. Remote PIN entry does not override that requirement or output readiness.

The page includes channel parameters, control mapping, mixing/preview, timer settings/actions/export, maps, network options, restart and firmware update. Use each section's Save/Apply action and read its response; changing a field alone is not proof it was saved. Finish an explicit setup session on the handset before attempting to enable live output.

13. Offline maps

The browser prepares a map package and transfers it to the handset. It is not a continuous map-streaming service to the LCD.

1. Open the web map selector and search an address/place.

2. Choose Street (OpenStreetMap) or Satellite (Esri), then pan/zoom to the area.

3. Check the **square download area**. Download selected area creates the browser preview; it has not yet saved to the handset.

4. Review the preview, name and attribution. Current image packages are limited to 512×512 pixels.

5. If storage is unavailable, use **Prepare map storage** and its confirmation; wait for an explicit result. Storage preparation is not map upload.

6. Choose **Save map to handset**. Wait for a success message, not just upload progress or "waiting for handset to validate and save".

7. Open the handset GPS page/widget to check the installed map.

Export map package saves a reusable package to your computer; Import accepts a supported package for transfer. Download handset map retrieves the installed one. A prepared browser preview and the installed handset map can be different.

Persistent storage holds the saved package; PSRAM is working memory used to draw it and is lost on power-off. A larger download can exhaust package/storage limits. If "Invalid setup session" appears, refresh/re-authenticate, confirm CUT, and retry; a stale browser session is not proof the map data is invalid.

Map credits must remain attached: street data © OpenStreetMap contributors; satellite imagery credits include Esri, Vantor, Earthstar Geographics and the GIS User Community as supplied by the imagery service. Search uses Photon. Preserve the package's actual source attribution when exporting/sharing.

14. Handset settings

Open **Setup > Handset** and page for additional entries.

Area	Main settings
Display	Screen timeout, idle action (logo saver/display off), brightness, colour scheme, activate now and orientation.
Sounds / alerts	Available sound and alarm options.
TX battery	Chemistry, cell count, low alarm and voltage sensing. The installed pack is 3S LiFe.
Network / web	Home Wi-Fi, auto-connect, setup access, web address/credentials and hostname.
System	Input diagnostics, firmware information, PIN and restart.

Orientation changes reflow the UI between portrait and landscape; they are a manual test/control, not accelerometer-driven rotation. Perform changes while CUT and finish any active editor/hold first. The installed default is landscape. The logo saver alternates JR PROPO and R-1 EVO. Touch or a navigation key wakes it; the wake action need not also operate the underlying control.

TX battery settings describe the sensed **raw pack feed**, even if the computer itself runs from USB or a separate 5V regulator. If that feed is absent, no TX battery reading can be inferred from the power source. Choose appropriate alarm settings for the actual chemistry rather than copying LiPo values into LiFe.

Restart reboots in CUT and clears RAM session data such as lap records. It does not deliberately erase saved models, maps or Wi-Fi. Export lap records first.

15. ExpressLRS module settings

Open **Setup > Model > RF module** with module communication approved and the Ranger powered. Discovery reads the module's own parameter tree. Available choices depend on its firmware; a missing option is not added by the handset.

Open folders, select a value, then Apply. Wait for readback before assuming the module accepted it. Typical advertised options include packet rate, telemetry ratio, Model Match, maximum power and Dynamic power. **Dynamic** lets the module vary transmit power according to its link logic, subject to the configured maximum; it is not a channel mixing or Walk-mode setting.

Command entries may require Run command then Confirm. Module Wi-Fi can interrupt the normal RF/CRSF link; reconnect and Refresh to establish its actual state. Binding phrase/UID support has its own workflow/version requirements. Numeric or string parameter types that the current editor does not support are read-only. The receiver needs compatible settings to reconnect after RF changes. See the RF configuration notes for protocol details.

ESC settings currently include preview/prototype functionality. Do not treat a visible menu entry as evidence of working ESC programming for a particular ESC.

16. Firmware, simulator and service tools

Firmware updates

Use the documented local build/test workflow for this installed XIAO/Waveshare board. USB upload uses PlatformIO and its hardware/branch preflight, including when a Pi is used only to receive the binaries and flash over USB. Do not substitute a diagnostic or different-board image for a normal update.

The web firmware updater accepts a targeted **.r1ota package**, produced by `tools/package_ota.py`, rather than an arbitrary binary. Authenticate, keep outputs CUT, choose the package and wait for validation/restart before reloading. Do not interrupt power. An end-to-end authenticated Wi-Fi update to **06b6928** was verified on 1 November 2026, including validation, reboot and web-interface availability. Saved models, maps and Wi-Fi settings were retained. Wi-Fi availability still depends on signal and network conditions.

The updater checks target, size, image header and transfer checksum before selecting the inactive application slot for restart in CUT. An interrupted or rejected upload leaves the current application selected. Web authentication is required; the checksum is not a cryptographic firmware signature. Use a trusted network. USB remains the recovery route and is required for bootloader or partition-layout changes. An update is not a substitute for checking model response before the next drive.

For the installed landscape profile, the guarded command-line updater uses `JR_R1_OTA_URL=http://HANDSET-IP` with `.venv/bin/pio run -e seeed_xiao_esp32s3-pcb -t upload`. Replace HANDSET-IP with the actual address. Enter the web password when prompted; do not publish it in a URL or script. This path retains the upload preflight.

Desktop/browser simulator

The simulator runs shared firmware UI code with simulated controls and telemetry. Use it to learn menus, explore layouts and test draft settings without hardware. Its model/state and sensor fixtures are separate from the handset and are not transferred automatically. Font rendering and display appearance may differ from the physical LCD. See Development guide for launch commands.

Both portrait (320 × 480) and landscape (480 × 320) layouts are maintained and checked in the simulator. **Setup > Handset > Display > Orientation** changes the display orientation. Automatic rotation is not implemented; it would require an accelerometer and orientation-detection firmware in a future revision.

Serial diagnostics

At 115200 baud with newline after commands, `BOOTLOG` retrieves recent startup results, `MEMORY` reports internal RAM/PSRAM and task stack margins, and `INPUTMON` opens the input monitor. `WIFISTATUS` reports network access details and may expose credentials; do not publish its unredacted output. Close a serial monitor before flashing if it holds the USB port.

A native screenshot export is available over USB/web rendering. It does not read back display glass, so a good capture cannot rule out LCD wiring or panel faults.

17. Troubleshooting

Symptom	Checks / action
Hold Drive has no effect or stays CUT	Open Output status. Check active setup session, module approval, neutral, fresh inputs/link and startup status. Use the larger Enable output control to check hold recognition.
Setup session active	Finish Wi-Fi setup on the handset. Closing the browser alone may leave the session active.
Link wait	Check receiver/module power, compatible binding/RF settings, module approval and fresh telemetry.
Screen works but TX voltage is missing	Check Voltage sensing On and the raw TX battery sense feed; USB/5V_SYS alone need not supply it.
Bars move while CUT, servo does not	Expected for requested-value widgets. Check Data sent values; enable only after readiness checks.
Wrong steering direction	Change the channel's Direction, then check neutral and both endpoints.
Input shows active with nothing connected	Inspect input monitor and PCB pull-ups/solder bridges. Calibration cannot repair a grounded digital input.
Web server times out	Check antenna/power and Wi-Fi status. Try the displayed numeric IP and refresh authentication. A Wi-Fi icon alone does not prove the server is reachable.
Invalid setup session on save	Refresh/authenticate to obtain a current session; remain CUT, then retry.
Map preview changed but handset map did not	Download prepares the browser preview; Save map to handset and success confirmation are separate.
Battery cells missing	Check actual sensor protocol/receiver forwarding and freshness. Expected cell count does not synthesise cells.
No GPS trail	Confirm receiver GPS forwarding, valid location/fix and fresh telemetry; a saved map alone provides none of these.
Timer press ignored	Check assignment/context, mode, minimum-lap interval and 100-lap capacity.
Laps disappeared after restart	Expected: lap history is session RAM. Export CSV before restarting.
Blank LCD but web/serial works	Separate backlight from graphics. Check LCD power, CS, reset, DC, clock and MOSI connections; touch uses I2C separately. Use the installed pinout guide.
Cannot exit Walk	Enter the correct PIN within the prompt; otherwise it times out back to Drive. Use USB recovery if the PIN is forgotten.

18. Settings and terminology reference

Term	Meaning
Requested	Control/mix value before safety substitutes a safe output.
Sent	Value actually submitted to the channel transport; can differ during CUT.
Role	Steering or throttle function assigned to a channel.
Assignment	Association of a physical control with an output, action or parameter.
Endpoint	Output travel limit; distinct from physical-input calibration.
CRSF	Handset-to-module channel/management/telemetry communication.
ExpressLRS	RF link firmware/system used by the Ranger and receiver.
S.Port	Sensor telemetry protocol supported by the project's custom receiver work.
LQ	Link-quality telemetry; not an output-enable switch.
PSRAM	Volatile working memory, not permanent map storage.

Scope of this edition: checked against transmitter firmware 06b6928. The installed hardware is the custom R-1 PCB with XIAO ESP32-S3 and Waveshare 3.5-inch RPi LCD (F). Receiver firmware/sensor capabilities are separate and should be checked for the actual receiver build. This manual makes no claim that every sensor, RF setting or prototype menu has completed hardware validation.

R-1 EVO is independent handset firmware, not an OpenTX/EdgeTX fork. ExpressLRS provides the radio link. JR PROPO's original R-1 hardware/manual, sourceTX and other radio interfaces informed the project; see the project README for credits and the separate build history for its development story.